

Data Warehouse Design Q&A

1. Highest Normal Form of Staging Database

The staging database is in First Normal Form (1NF) because all fields contain atomic values and there are no repeating groups.

However, it is not fully normalized to higher forms (2NF/3NF) because there are redundancies (e.g., repeated product and store details in the same dataset).

2. Difference Between Dimension Table & Fact Table

A **dimension table** stores descriptive attributes (e.g., ProductName, StoreName, Province) and is used for filtering and grouping data in analysis.

A **fact table** stores measurable business data (e.g., sales amount, quantity sold) and contains foreign keys that link to dimension tables.

3. Tables After SSIS Process (Data Warehouse Design)

After the ETL process, the data is organized into a **star schema** with the following tables:

- **DimProduct** → Stores product details (ProductKey, ProductName, Category, Price)
- **DimStore** → Stores store information (StoreKey, StoreName, Province)
- **DimDate** → Stores date-related attributes (DateKey, FullDate, Year, Month, Day)
- **FactSales** → Stores transactional data (DateKey, StoreKey, ProductKey, QuantitySold, TotalSales)

This structure reduces redundancy and improves query performance for reporting tools like Power BI.

4. Entity-Relationship Diagram (ERD) Explanation

The data warehouse follows a **star schema design**, where:

- **FactSales** is the central table
- It connects to:
 - **DimProduct** via ProductKey
 - **DimStore** via StoreKey
 - **DimDate** via DateKey

Each dimension table has a **primary key**, and FactSales contains these as **foreign keys**.

Relationship Summary

- One **Product** → Many **Sales**
- One **Store** → Many **Sales**
- One **Date** → Many **Sales**

ERD From Visual Studio

